

CEAHLAU TOACA, Romania

WMO: 151080

Lat: 46.9775N

Lon: 25.9500E

Elev: 6229

StdP: 11.68

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.3	(c) 0.7	(d) -13.5	(e) 3.3	(f) 19.9	(g) -6.7	(h) 4.8	(i) 11.4	(j) 77.0	(k) 19.6	(l) 76.4	(m) 19.5	(n) 20.4	(o) 50	(p) 0.948

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 11.5	(c) 67.9	(d) 59.6	(e) 64.9	(f) 58.2	(g) 62.2	(h) 56.5	(i) 62.0	(j) 65.5	(k) 59.7	(l) 62.9	(m) 57.6	(n) 60.6	(o) 9.0	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.9	100.9	63.7	58.5	92.6	61.3	56.5	86.0	59.3	31.2	65.7	29.4	62.8	27.8	60.7	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 75.7	(b) 62.3	(c) 45.4	(e) -7.8	(f) 72.6	(g) 4.1	(h) 3.4	(i) -10.8	(j) 75.0	(k) -13.2	(l) 77.0	(m) -15.5	(n) 78.9	(o) -18.5	(p) 81.3		
(a) 75.7	(b) 62.3	(c) 45.4	(e) -7.9	(f) 64.9	(g) 4.1	(h) 3.3	(i) -10.9	(j) 67.3	(k) -13.3	(l) 69.2	(m) -15.7	(n) 71.1	(o) -18.7	(p) 73.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.0	18.0	18.8	22.4	32.7	42.6	48.9	52.1	52.5	44.0	36.5	29.3	20.9		
	DBStd	15.11	9.41	9.08	8.83	8.30	7.75	6.74	6.40	6.69	7.93	9.58	10.25	9.50		
	HDD50	5855	993	873	856	519	250	100	53	51	211	426	620	903		
	HDD65	10961	1458	1293	1321	968	696	483	400	390	631	883	1070	1368		
	CDD50	371	0	0	0	1	19	67	118	128	30	8	0	0		
	CDD65	3	0	0	0	0	0	0	1	2	0	0	0	0		
	CDH74	1	0	0	0	0	0	0	0	1	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	17.1	21.9	22.1	19.6	15.4	13.7	12.3	14.1	12.2	14.8	16.0	21.3	22.5		
	PrecAvg	25.6	0.8	0.9	1.3	2.0	3.0	4.3	4.0	3.3	2.0	1.7	1.1	1.2		
	PrecMax	40.4	1.6	1.7	2.6	3.9	6.3	11.1	7.9	6.1	5.1	4.3	2.2	2.8		
	PrecMin	18.5	0.3	0.3	0.5	0.6	1.3	1.5	1.7	1.1	0.7	0.2	0.0	0.1		
	PrecStd	4.5	0.3	0.3	0.5	0.9	1.4	2.1	1.6	1.4	1.0	0.9	0.6	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.4	43.2	45.0	56.5	63.5	68.9	71.2	72.2	66.6	60.6	52.2	43.6		
		MCWB	33.1	34.6	37.0	47.6	54.8	60.3	62.1	61.5	54.7	50.2	42.2	33.8		
	2%	DB	35.5	36.3	40.6	50.7	59.1	64.7	67.5	68.0	61.2	55.6	49.1	39.0		
		MCWB	31.8	32.8	36.1	44.0	52.6	58.2	60.5	59.4	53.6	47.1	40.0	32.9		
	5%	DB	32.7	33.1	37.1	46.7	55.8	61.2	64.4	64.8	57.6	52.1	45.9	36.0		
		MCWB	30.5	30.7	33.8	41.6	50.8	56.0	58.9	58.0	52.2	45.2	38.7	31.9		
	10%	DB	30.4	30.7	34.0	43.2	53.0	58.5	61.3	61.8	54.4	48.9	43.1	33.2		
		MCWB	28.6	29.2	32.1	39.8	48.9	54.5	57.0	56.3	50.3	44.3	37.2	30.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.3	37.5	40.3	50.4	57.9	63.6	65.1	65.0	58.1	54.8	46.6	38.4		
		MCDB	37.0	38.8	42.3	54.3	60.2	66.3	68.0	68.8	61.9	58.4	49.2	39.7		
	2%	WB	33.1	33.5	37.0	45.2	54.2	59.8	62.1	61.8	55.3	50.1	43.3	35.7		
		MCDB	34.1	35.7	39.4	48.7	57.2	62.4	65.5	65.6	59.3	52.5	46.0	37.0		
	5%	WB	31.2	31.6	34.4	42.6	51.7	57.2	59.9	59.3	52.9	47.4	41.1	33.2		
		MCDB	32.2	33.0	36.3	45.6	54.6	60.1	63.1	62.6	56.4	49.8	43.8	34.3		
	10%	WB	28.8	29.4	32.2	40.2	49.6	55.0	57.7	57.1	50.7	45.0	38.8	31.2		
		MCDB	29.7	30.3	33.7	42.8	52.2	57.7	60.4	60.5	53.8	47.7	41.4	32.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.4	9.0	9.7	10.1	11.5	11.7	11.6	11.5	10.4	9.5	8.8	8.4		
		MCDBR	8.9	10.5	12.2	13.8	14.6	15.2	15.1	14.1	13.6	11.8	10.3	9.1		
		MCWBR	7.6	9.1	9.5	10.0	10.6	11.1	11.2	10.4	10.4	9.3	7.7	7.0		
	5% WB	MCDBR	8.2	9.9	11.0	12.7	13.9	14.4	13.7	13.6	12.8	11.2	9.7	8.1		
		MCWBR	7.8	9.1	9.2	10.0	11.3	11.6	11.5	11.2	10.7	9.6	8.6	7.5		
(40) Clear-Sky Solar Irradiance	taub		0.243	0.265	0.296	0.339	0.325	0.324	0.340	0.345	0.314	0.287	0.267	0.245		
	taud		2.455	2.425	2.430	2.360	2.441	2.456	2.403	2.389	2.466	2.542	2.550	2.495		
	Ebn at Noon		279	292	297	291	298	298	291	285	286	280	267	265		
	Edh at Noon		20	26	30	36	35	34	36	35	29	23	19	17		
(44) All-Sky Solar Radiation	RadAvg		393	649	1012	1372	1719	1826	1824	1669	1185	785	428	304		
	RadStd		55	79	115	130	156	156	154	140	137	99	49	38		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.46	N/A	N/A	N/A	N/A	N/A	-437	-545	+92	N/A
(47) Regional (55 neighbors)		+1.24	N/A	N/A	+1.48	N/A	N/A	-284	-396	+149	+34

Nomenclature: See separate page